Abstract #02290 ISMRM ABSTRACT

Fully Bayesian Uncertainty-Aware Biomarkers for Image-Based Cancer Diagnosis: Application to Prostate Diffusion MRI

Primary: Analysis Methods - Classification and Prediction **Secondary:** Body - Prostate **Keywords:** MACHINE LEARNING/ARTIFICIAL INTELLIGENCE DIFFUSION-WEIGHTED IMAGING PROSTATE MRI BAYESIAN INFERENCE UNCERTAINTY ESTIMATION

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Impact

Diffusivity spectra derived via Bayesian analysis of prostate water diffusion decays contain detailed tissue-specific information. Logistic regression identifies which spectral diffusivity components contribute most to tissue classification. The concurrent classification probability helps weigh the significance of findings in clinical settings.

Synopsis

Motivation: Clinical prostate cancer detection relies on diffusion-weighted and T2-weighted MRI with ADC as a quantitative biomarker. Diffusivity spectra from multi-b scans may provide more reliable biomarkers reflecting microscopic tissue changes.

Goals: Apply Bayesian diffusivity spectral analysis to multi-b data and derive uncertainty-aware tissue classifiers.

Approach: Diffusivity spectra were estimated from multi-b diffusion-weighted data in patients with suspected prostate tumors using Bayesian inference. Logistic regression based on spectral features was compared to conventional ADC-based classification.

Results: Characteristic spectra were derived for normal/tumor regions (peripheral/transition zones). Spectra-based classification matched closely ADC performance for tumor detection and showed slightly superior performance for Gleason Grade Group classification.

Introduction

Bayesian spectra reconstruction, which expands on compartmentalization [1,2], has been described previously in [3,4]. The present work extends this earlier work significantly by jointly estimating spectra *and SNR* with an improved sampler and is thus *fully Bayesian.* We apply the method to a larger cohort of patient data and perform logistic regression for tissue classification.

Data

Data analysis is performed on an existing prostate dataset, previously described in [1], comprising 56 patients with suspected prostate cancer. Multi-parametric MRI, including diffusion-weighted images, was acquired on a GE 3T Discovery MR750 scanner with an endorectal coil. The diffusion scan protocol included 15 b-values from b=0 to b=3500 s/mm² in uniformly spaced 250 s/mm² increments.

Our present analysis includes 149 manually-annotated regions of interest (ROIs): 109 normal tissue and 40 tumor regions across the peripheral zone (PZ: 27 tumor, 54 normal) and transition zone (TZ: 13 tumor, 55 normal), as delineated by an experienced radiologist. Gleason Grade Groups (GGG), derived from histopathology, were available for 28 PZ tumor ROIs.

Methods

Probabilistic Modeling

Following the Stejskal-Tanner model, the observed signal decay $s \in \mathbb{R}^m_{\geq 0}$ is modeled as a Laplace transform over a discrete set of diffusivity components $D \in \mathbb{R}^n_{>0}$ with corresponding relative fractions $R \in \mathbb{R}^n_{>0}$: $s = UR$ where $U_{ji} = e^{-b_j D_i}$. Under Gaussian IID noise: $p(s|R,\sigma) = \mathcal{N}(s; UR, \sigma^2 I)$. We jointly infer spectrum R and noise σ via hierarchical Bayesian modeling: $R_i \sim \mathrm{HalfNormal}(0, \lambda^{-1})$ (ridge regression), $\sigma \sim \mathrm{HalfCauchy}(\beta)$ (robust scale prior [6]). The joint posterior is $p(R,\sigma|s) \propto p(s|R,\sigma)\, p(R)\, p(\sigma)$.

Approximate Inference with Uncertainty Quantification

We sample using the No-U-Turn Sampler (NUTS) [7], which efficiently explores the joint posterior via gradient-based Hamiltonian Monte Carlo with automatic tuning. This eliminates manual SNR estimation and adapts to actual scanner noise properties. We use Bootstrap [8] to propagate uncertainty through the subsequent logistic regression.

Constructing Biomarkers

To predict prostate cancer diagnosis and grade, we train L2-regularized logistic regression classifiers using Leave-One-Out Cross-Validation (LOOCV). Features include individual diffusivity fractions ($D \in \{0.25, 0.5, ..., 3.0, 20.0\}$ µm²/ms). The D=20 µm²/ms compartment represents the intravascular component [13]. We evaluate AUC with bootstrap CIs (95%) for tumor vs. normal classification (stratified by prostate zone) and Gleason grade stratification, comparing against monoexponential ADC baselines (b=0-1250 s/mm²) and individual spectra features.

Results

Inference Configuration

NUTS sampling (2000 iterations with warmup of 200) with ridge prior (λ=0.1) and HalfCauchy noise prior (β=1.0) yielded well-converged posteriors for all 149 ROIs (see Fig.1 top, Fig.2).

Classification Performance

L2-regularized logistic regression (C=1.0, LOOCV) with top-5 feature selection achieved competitive tumor detection: PZ (AUC=0.94 [0.87-0.99]), TZ (AUC=0.93 [0.81-1.00]), matching monoexponential ADC baselines (0.94, 0.96 respectively). For intermediate GGG stratification (1-2 vs 3-5), our

approach (AUC=0.82 [0.63-0.96]) slightly outperformed ADC (0.79 [0.58-0.96]) despite small sample size (n=28), see Fig.3.

Uncertainty & Feature Importance

Monte Carlo propagation (N=200 samples) provided calibrated epistemic uncertainty: predictions with uncertainty intervals crossing the 0.5 threshold showed 1.98× higher misclassification rates (Fig.4, Fig.5), enabling confidence-aware clinical decisions.

Discussion and Conclusion

Spectra-derived biomarkers achieve ADC-competitive tumor detection while providing interpretable microstructural features. The ADC findings align with prior studies [9-13].

Data quality in terms of SNR and absence of motion-related signal loss is sufficient to reliably derive spectral data for all cases. Our NUTS-based approach [7] eliminates manual SNR tuning at the expense of computation. Tissue classification ability varies widely for the different diffusion bins, but the lowest derived diffusivity (0.25 μm²/ms) consistently results in high discrimination. Notably is the tumor detection ability of the free diffusion bin (3.0 μm²/ms), particularly for the PZ. This may be explained by the replacement of lacunar spaces (free diffusion) with tumor tissue (restricted diffusion). For Gleason grade classification, bins with reduced diffusion (1.5-2.0 μm²/ms) seem to be important, since they likely represent solid tissue with larger benign cells. Decreased perfusion (20.0 μm²/ms) seems to be an indicator of malignancy. The logistic regression coefficients convey similar information, i.e., they clearly reveal which diffusion characteristics and related microscopic features causing these changes result in most reliable classification. One added benefit of logistic regression analysis is the ability to provide a classification probability, critical for clinical translation. The small dataset (149 ROIs, 40 tumors) limits generalizability. Further protocol optimization is needed to obtain suitable high-b diffusion data in larger patient cohorts without using endorectal coils. Real-time inference with neural network-based approaches might help with clinical translation. The integration of T2-mapping, automated segmentation, and texture analysis might further improve grade stratification.

Acknowledgements

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Figures and Tables

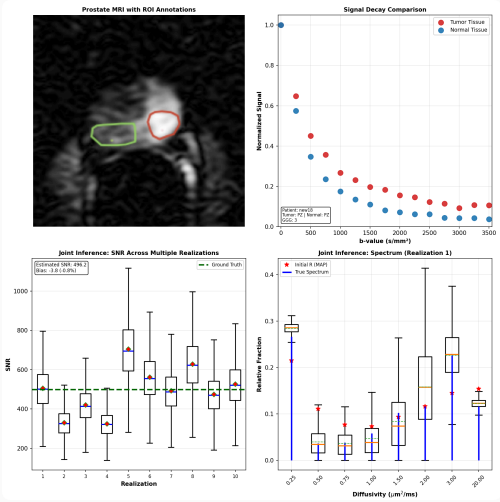

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Figure 1: *Top: Prostate MRI at b=3500 s/mm² (4.4x4.4x5 mm³ resolution) with signal decay comparison showing tumor vs normal tissue from patient ROI. Bottom: Joint SNR-spectrum inference validation using NUTS on synthetic data (n=10 noise realizations, 8,000 MCMC samples). Left: SNR posteriors across realizations; ground truth=500 (green), -0.8% bias. Right: Spectrum posterior (realization 1); true spectrum (blue), MAP init (red). Hierarchical Bayesian framework simultaneously recovers signal and noise.*

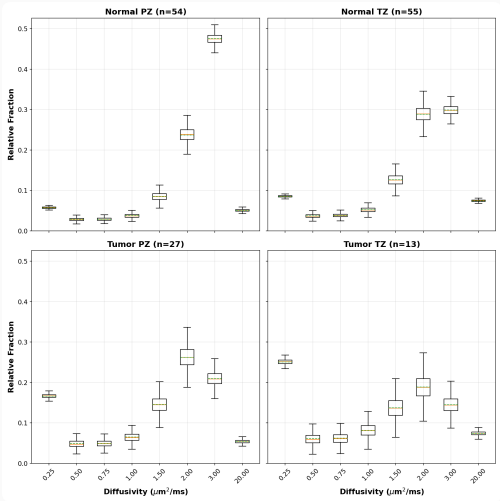

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Figure 2: *Averaged diffusivity spectra across patients for different tissue types show distinct distributions in normal and tumor regions in both peripheral (PZ) and transition (TZ) zones. Boxplots represent posterior distributions from NUTS inference with ridge prior (λ=0.1). Tumor regions exhibit elevated low-diffusivity fractions (0.25 μm²/ms) consistent with restricted diffusion from dense cellularity. N=109 normal, N=40 tumor ROIs.*

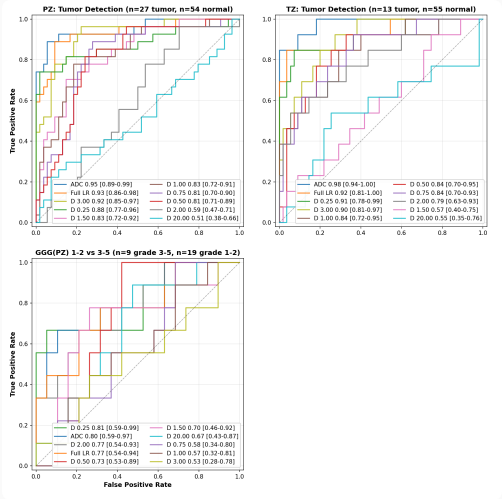

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Figure 3:

ROC curves for three classification tasks using leave-one-out cross-validation (LOOCV). Top left: PZ tumor detection (N=81, AUC=0.94 [0.87-0.99]). Top right: TZ tumor detection (N=68, AUC=0.93 [0.81-1.00]). Bottom: Gleason grade stratification for PZ: GGG 1-2 vs 3-5 (N=28, AUC=0.82 [0.63-0.96]). Individual diffusivity bins (D) and full logistic regression (Full LR) shown. ADC (b = 0-1250 s/mm²) baseline included for comparison.

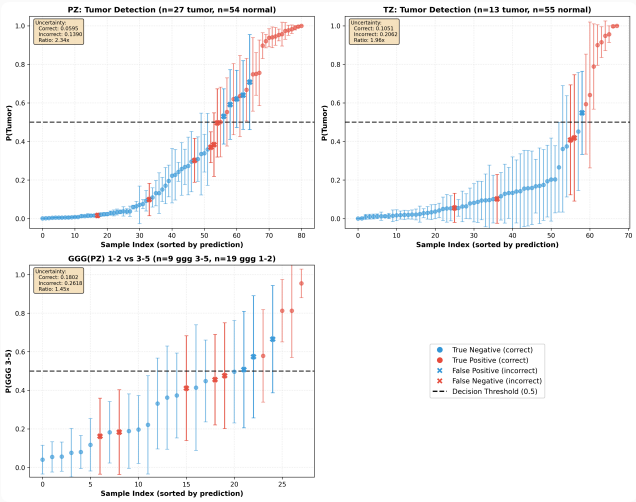

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Figure 4:

Uncertainty propagation in prostate cancer classification. Bootstrap uncertainty quantification (n=200 posterior samples) for logistic regression predictions across three clinical tasks. Error bars represent prediction uncertainty; samples sorted by predicted probability. Misclassified samples exhibit significantly higher uncertainty (1.5-2x), demonstrating the method's ability to identify ambiguous cases requiring clinical review.

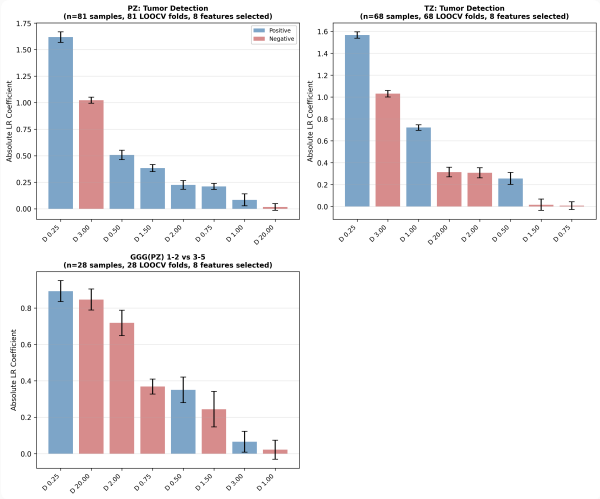

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Figure 5:

Logistic regression coefficients sorted by magnitude reveal discriminative diffusivity patterns. Low diffusivity (0.25 μm²/ms) shows strongest positive association with tumor across PZ, TZ, and GGG (PZ) tasks, while intermediate diffusivity (3.0 μm²/ms) exhibits negative association. Grade stratification shows distinct pattern with a strong negative coefficient at 20.0 and 2.0 μm²/ms. Color indicates sign: blue for positive, red for negative.